Uranium

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This silvery-gray element changed the course of history. Its impact on the world, both in terms of its use as a nuclear fuel and in nuclear weapons, has been enormous. It was formally discovered by German chemist Martin Heinrich Klaproth, but arguably the discovery of radioactivity by the Nobel Prize–winning French physicist Henri Becquerel, through his study of uranium, opened the door to the nuclear age.

Chunk of pure uranium

More than

8 million tons

of uranium can be found in Earth's crust.

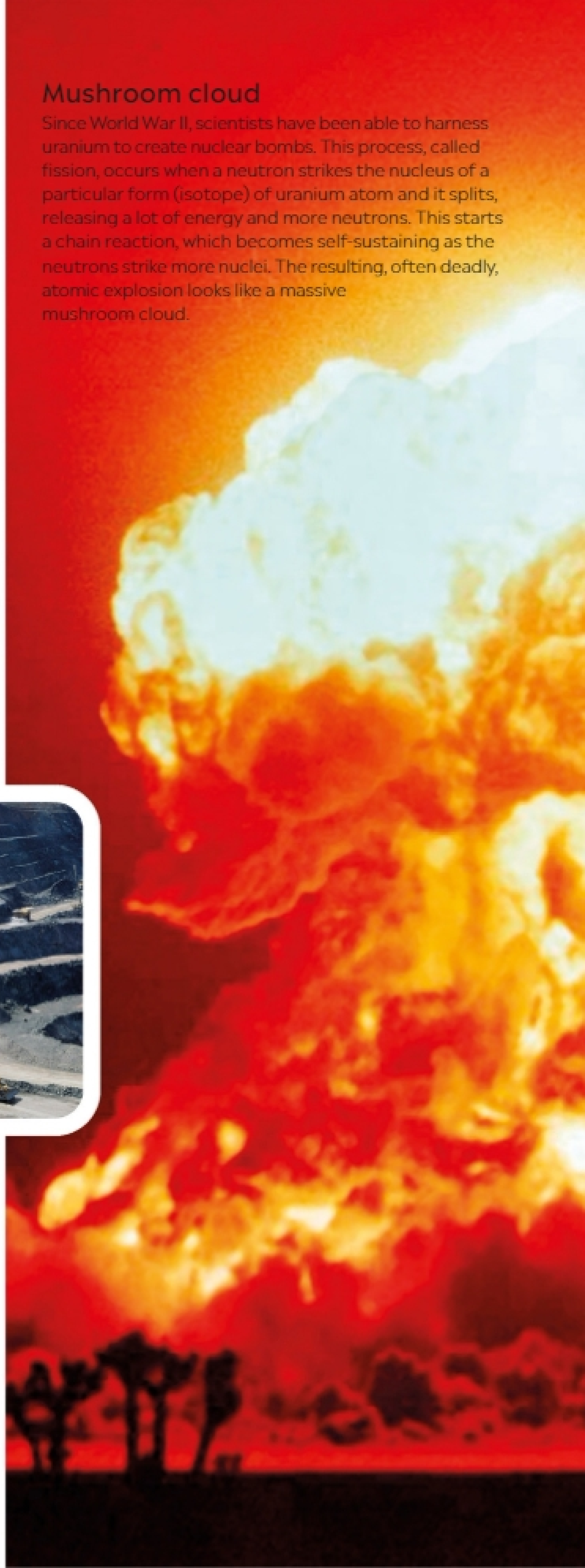
Mining uranium

The Ranger uranium mine in the Northern Territory of Australia (right) is one of the world's largest sources of uranium. Once uranium ore is extracted from the ground, a series of chemical reactions are used to turn it into uranium dioxide. This is then used to make nuclear fuel rods for nuclear reactors.

This glaze contains up to 14 percent uranium.

Radioactive red

Fiestaware tableware was particularly popular in the 1930s. The red glazes that were applied to the plates, cups, and saucers included a significant amount of uranium oxide, which made them radioactive. The glaze was eventually discontinued, although non-radioactive Fiesta ceramics are still made.



Mushroom cloud

Since World War II, scientists have been able to harness uranium to create nuclear bombs. This process, called fission, occurs when a neutron strikes the nucleus of a particular form (isotope) of uranium atom and it splits, releasing a lot of energy and more neutrons. This starts a chain reaction, which becomes self-sustaining as the neutrons strike more nuclei. The resulting, often deadly, atomic explosion looks like a massive mushroom cloud.